

UNIT 1



FUNDAMENTALS OF COMPUTER



Q.1: WHAT IS COMPUTER?

A computer is an electronic data processing machine or device that performs calculations and operations based on instructions provided by a program.

Q.2: DISCUSS THE USE OF COMPUTER IN ANY TWO FIELDS OF LIFE

Computers are now used vastly in every field of life. The advancement in computer caused advancement in knowledge, science and technology. Today, computers are used in fields of business, industry, education, banking, transportation, research, explorations, media, entertainment, etc.

USE OF COMPUTERS IN EDUCATION

Computer technology has had a deep impact on the education. Because of computers, education has become easier and much more interesting than before.

THE MAIN USES OF COMPUTER IN EDUCATION ARE AS FOLLOWS:

- (i)** Computer teaching plays a key role in the modern education system. Students find it easier to refer to the Internet than searching for information in fat books. The process of learning has gone beyond learning from prescribed textbooks. Internet is a much larger and easier-to-access storehouse of information.
- (ii)** When it comes to storing retrieved information, it is easier done on computers than maintaining hand-written notes.
- (iii)** Computer technology has made the dream of distance learning, a reality. Education is no longer limited to classrooms. It has reached far and wide.
- (iv)** Computers facilitate effective presentation of information. Computers facilitate audio-visual representation of information, thus making the process of learning interactive and interesting. Computer-aided teaching adds a fun element to education.
- (v)** The Internet can be used to refer to information on different subjects. Both teachers and students benefit from the Internet. Teachers can refer to it for additional information and references on the topics to be taught.
- (vi)** Computers enable storage of data in the electronic format, thereby saving paper. Memory capacities of computer storage devices enable them to store huge chunks of data.

USE OF COMPUTERS IN MEDICAL SECTOR:

Computers are the excellent means for storage of patient related data. Big hospitals employ computer systems to maintain patient records. It is often necessary to maintain detailed records of the medical history of patients. Doctors often require the information about a patient's family history, physical ailments, already diagnosed diseases and prescribed medicines.

Medical journals, research and diagnosis papers, important reference books can best be stored in

an electronic format. Documents and Many of the modern day medical equipment have small, programmed computers.

Many of the medical appliances of today work on pre-programmed instructions. The circuitry and logic in most of the medical equipment is basically a computer.

Computer software is used for diagnosis of diseases. It can be used for the examination of internal organs of the body. Advanced computer-based systems are used to examine delicate organs of the body. Some of the complex surgeries can be performed with the aid of computers. The different types of monitoring equipment in hospitals are often based on computer programming.

Medical imaging is a vast that deals with the techniques to create images of the human body for medical purposes. Many of the modern methods of scanning and imaging are largely based on the computer technology. Magnetic resonance imaging and tomography employs computer software.

In the field of medicine, computers allow for faster communication between a patient and a doctor. Doctors can collaborate better over the Internet. Today, it is possible to obtain experts opinions within seconds by means of the Internet.

Medical practitioners can discuss medical issues in medical forums. They can exchange images and messages in seconds and derive conclusions speedily. They can seek advice and share knowledge in a convenient manner over the Internet. The importance of computers cannot be stressed enough as computer technology has revolutionized the field of medicine.

Q.3: DESCRIBE THE MECHANICAL ERA OF COMPUTER EVOLUTION

ELECTRO-MECHANICAL ERA (DARK AGE):

Men have been trying to invent machines that can solve mathematical problems. In mechanical era, human became successful to make simple machines that could help performing simple arithmetic operation, in other words computing. These machines were manually operated since the electricity was not invented. Some of the important machines of this era are:

(A) ABACUS (3000 B.C.):

Abacus was invented about 5000 years ago. It is also known as counting frame. Abacus is still used to teach basic arithmetic operations to the students. Abacus is considered as first computer prototype.

(B) NAPIER'S BONES (1612 A.D.):

Scottish mathematician John Napier developed Napier's Bones, in 1612. It is also called Napier's Rods. It was a small machine that contained ten rods. The rods were made up of bones, wood or metal. Napier's Bones was used for multiplication and division.

(C) SLIDE RULE (1622 A.D.):

English mathematician William Oughtred invented a Slide Rule. This was basically a pair of two moveable rules placed side by side. They had a standardized set of markings called scales. These scales were used for mathematical computations. Slide Rule was used to multiply and divide numbers. The modern slide rules are still countries.

(D) PASCALINE OR PASCAL'S CALCULATOR (1642 A.D.):

French mathematician Blaise Pascal invented a mechanical calculator which was called Pascaline.

He developed this machine to help his father who was a tax collector. Pascaline can perform addition and subtraction by eight figures. User had to adjust wheels in order to perform calculations.

Q.4: DESCRIBE THE MECHANICAL ERA OF COMPUTER EVOLUTION**ELECTRO-MECHANICAL ERA (MIDDLE AGE):**

This era starts from the mid of 19th century. In this era scientists became able to develop faster and more accurate computing machines as they were powered by steam and electricity. One of such machines was Tabulating Machine.

HERMAN HOLLERITH'S TABULATING MACHINE (1890 A.D.):

It was invented by American inventor Herman Hollerith in 1890. Hollerith's first tabulator was used for the U.S. 1890 Census. Because of Hollerith's tabulating machine census data took only six months to compile, which was very fast as compared to previous U.S. census in 1880 which took 7 years to be completed. Hollerith Tabulating Machine became very famous and used in many offices of U.S. Government. His tabulating machine could read and sorted out data.

Q.5: DESCRIBE THE ELECTRO ERA OF COMPUTER EVOLUTION.**ELECTRONIC ERA (MODERN AGE):**

The advancement in the vacuum or electron tube gave birth to the electronic era. In this era the true computers were invented which worked on the principle of input, store, process and output. Continuous advancement in electronic engineering increased efficiency and speed of computers considerably. The electronic era may be categorized in generations on the basis of core technology used to build computers.

- The First Generation of Computers (1940 to 1956)
- The Second Generation of Computers (1956 to 1963)
- The Third Generation of computers (1964 to 1971)
- The Fourth Generation of computers (1971 to Present)
- The Fifth Generation of Computers (Present and Beyond)

Q.6: DESCRIBE THE CHARACTERISTICS OF THE FIRST GENERATION OF COMPUTERS AND ITS MAIN FEATURES.**FIRST GENERATION OF COMPUTERS (1940 TO 1956):**

Based on vacuum tubes, first generation computers were very large in size. This generation used machine language (i.e. 1's and 0's). Magnetic drums were used as primary internal storage medium and punched cards for input. In this generation mainly batch processing operating system was used. Computers of this generation were primarily used for scientist and research purposes. Electronic Numerical Integrator and Calculator (ENIAC), Universal Automatic Computer (UNIVAC) are examples of first computers.

MAIN FEATURES:-

- Technology Used: Vacuum Tubes
- Size: Very large in size (Main frame)
- Language: Low level Machine language (i.e. 1's and 0's).
- Main memory (primary internal storage): Magnetic
- External storage: Punched card
- I/O Device: Punched cards and papers
- Operating system: Batch processing operating system, no proper operating system)
- Use: Scientific and research purpose.
- Example: ENIAC, EDVAC, UNIVAC

Q.7: WHAT DO YOU KNOW ABOUT SECOND GENERATION OF COMPUTERS AND ITS MAIN FEATURES?**SECOND GENERATION OF COMPUTERS (1956 TO 1963):**

Because of transistors computers became smaller, faster, cheaper and more efficient. Assembly language and a high-level language FORTRAN were introduced. Magnetic core was used as primary internal storage medium.

Punched Cards were used for input. Batch processing and Multiprogramming Operating systems

were used. These computers were mainly used for commercial productions, scientific and engineering analysis and design. Examples are IBM 7094 and IBM 1401. Main Features

- Technology Used: Transistors.
 - Size: smaller and faster (main frame), cheaper and more efficient.
 - Language: Assembly language and a high language FORTRAN
 - Main memory (internal storage): Magnetic core (RAM and ROM)
 - External storage: Magnetic tapes and magnetic disk
 - I/O Device: Punched card
 - Operating system: Batch processing and Multi-programming
 - Use: commercial productions, scientific and engineering analysis and design •
- Example: IBM \
- 7094 and IBM 1401

Q.8: BRIEFLY DESCRIBE THE THIRD GENERATION OF COMPUTERS AND ITS MAIN FEATURES**THIRD GENERATION OF COMPUTERS (1964 TO 1971):**

Use of ICs further decreased size of computers and increased the speed and efficiency. Less expensive computers were introduced. High level programming languages such as Pascal and COBOL were used. Keyboard as input and monitor as output also eased the use of computer. Time sharing and Real-time Operating Systems were used. The use of computers was extended to database management and automatic industrial control. IBM 360 and IBM 370 are the examples of this generation's computers.

Main Features

- Technology Used: Integrated Circuits (ICs).
- Size: Mini (decreased size and increased the and efficiency, less expensive)
- Language: High level programming language such as Pascal and COBOL
- Main memory (internal storage): PROM and DRAM
- External storage: Floppy disk
- I/O Device: Keyboard (input), Monitor (output)
- Operating system: Time sharing and Realtime (Unix operating system)
- Use: The use of computers was extended to database management and automatic industrial control
- Example: IBM 360 and IBM 370

Q.9: DESCRIBE CHARACTERISTICS OF THE FOURTH GENERATION OF COMPUTERS**FOURTH GENERATION OF COMPUTERS (1971 TO PRESENT):**

The invention of microprocessors was revolutionary which caused the development of faster, less expensive, smaller and more reliable computers. They used semi-conductor memories RAM and ROM and magnetic storage became popular. More high-level languages were introduced like C, C++, Java, etc. These computers are used in almost every field of life, like space applications, business and art work.

Time sharing, real time and distributed operating system are used. This generation also saw the development of graphics User Interfaces (GUIS). Examples are Apple Macintosh, IBM PC.

Main Features

- Technology Used: Microprocessors.
- Size: microcomputer (faster, less expensive, smaller and more reliable)
- Language: More high languages like C, C++, Java etc.
- Main memory (internal storage): Semi-conductor memories RAM and ROM (EPROM and SRAM)
- External storage: Magnetic storage
- I/O Device: Monitor (output)
- Operating system: Time sharing, real time and distributed operating development of Graphical User Interfaces (GUIS) (DOS and UNIX)
- Use: in every field of life like space applications, business and art
- Example: Apple Macintosh, IBM PC.

Q.10: WHAT DO YOU KNOW ABOUT THE FIFTH GENERATION OF COMPUTER**FIFTH GENERATION OF COMPUTERS (PRESENT AND BEYOND):**

Fifth Generation computing devices are still being developed. In this generation, computers will be capable of self-learning, reasoning and generalization. These computers or controlled machines like robot will also be able to process human languages. The branches of AT include: Machine Learning, Deep Learning, Natural Language Processing, Robotics and Expert S

Main Features

- **Technology Used:** Artificial Intelligence Technology, still being developed • **Size:** very tiny in size.
- **Language:** AI (Artificial Intelligence) include; Machine Learning, Deep Learning. Natural Language Processing, Robotics and Expert Systems. These computers or controlled machines like robot will also be able to process human languages.
- **Main memory (internal storage):** EEPROM, SIMM, DIMM External storage: Modified magnetic and optical disk etc.
- **I/O Device:** Keyboard, pointing device, scanner (input) Monitor, printer (output) etc. Operating system: GUI based, Microsoft windows 95 and onward, Windows NT. Use: In all fields of life. Computers will be capable of self-learning, reasoning and generalization.
- **Example:** laptop, notebook, digital diary, Android, Pocket PC, Palm top, Humanoid Robots

Q.11: DESCRIBE ARTIFICIAL INTELLIGENCE WITH EXAMPLES.**ARTIFICIAL INTELLIGENCE (AI):**

Artificial intelligence (AI) is wide-ranging branch of computer science concerned with building smart machines capable of performing tasks that typically require human intelligence. AI is an interdisciplinary science with multiple approaches, but advancements in machine learning and deep learning are creating a paradigm shift in virtually every sector of the tech industry. There are many different fields of AI, including 'robotics', but one of the most commonly known forms is referred to as 'machine learning'. This involves a program applying known information to new experiences and 'learning' how to take this historical information and its experiences into account in future actions. Machine learning can find patterns in large amounts of data that humans might otherwise miss.

EXAMPLES OF ARTIFICIAL INTELLIGENCE:

(i) A Bedford, Mass. company made smarter home robot Roomba 980. The Roomba 980 model uses artificial intelligence to scan room size, identify obstacles and remember the most efficient routes for cleaning. The self-deploying Roomba can also determine how much vacuuming there is to do based on a room's size, and it needs no human assistance to clean floors.

(ii) Hong Kong based company made humanoid robot, Sophia. Sophia is an incredibly advanced social-learning robot. Through AT, Sophia can efficiently communicate with natural language and use facial expressions to convey human-like emotions. (iii) A Boston based company made AT powered machine for pathologists. The company's machine learning algorithms help pathologists analyze tissue samples and make more accurate diagnoses. The aim is to not only improve diagnostic accuracy, but also treatment. Path AI's technology can also identify optimal clinical trial participants. Path AI has worked with the Bill & Melinda Gates Foundation and Philips to develop high-volume prognostic test support tools and plans for sustainable access to their advanced diagnostic services.

USES OF ARTIFICIAL INTELLIGENCE:

- A.I. is also used for Machine Learning. It learns from our daily routines and suggests us different options. Like google maps suggest the best ways for our daily commute.
- A.I. is vastly used in scientific experiment, healthcare and space technologies.

Q.12: DESCRIBE THE CLASSIFICATION OF COMPUTERS ACCORDING TO TECHNOLOGY.**CLASSIFICATION OF COMPUTERS ACCORDING TO TECHNOLOGY:**

According to technology or type of data they handle, computers are classified into three types.

(a) Analog Computers:

Analog Computers are used to process analog data. Analog data are in the form of continuously varying physical quantities like pressure, temperature, voltage, speed and weight.

Examples of Analog computer are speedometer of a car, voltmeter etc.

(b) Digital Computers:

Digital Computers are most commonly used type of computers. They are used to process information with quantities using the binary number system (0's and 1's). Digital Computers are used in home, educational institutes, offices, business, scientific fields, etc.

(c) Hybrid Computers:

Hybrid Computers are the combination of Analog and Digital Computer system. These computers combine analog and digital features of computers in a single machine. A Hybrid Computer uses analog to digital and digital to analog conversion. It may input or output either digital or analog data.

- Use: in every field of life like space applications, business and art
- Example: Apple Macintosh, IBM PC.

Q.13: HOW CAN WE CLASSIFY COMPUTERS ACCORDING TO THEIR SIZES?**CLASSIFICATION OF COMPUTERS ACCORDING TO SIZE:**

Computers are also divided into four groups according to their size.

(a) Super Computers:

Super Computers are the most powerful, fastest and largest computers. They are extremely expensive. These computers are widely used in scientific applications such as aerodynamics, design simulations, processing of geological data, weather forecast nuclear research.

(b) Mainframe Computers:

Mainframe Computers are powerful multi-user and multi-processors computers. They can process huge amount of calculations at very high speed. Mainframes are also very expensive and require a lot of technical expertise to be installed and operated. They are used in banks and many large business organizations where several users work simultaneously.

(c) Minicomputer:

These are smaller than mainframe computers, but they are more powerful than microcomputers. Minicomputers usually use multi-user operating System. Multiple users can use the minicomputers through terminals. Minicomputers may be used as network servers and internet servers. DEC VAX and IBM AS/400 are good examples of minicomputers.

(d) Microcomputer

Microcomputers are also called Personal Computers (Pcs). The use of microprocessor has made computers cheaper yet faster and more reliable. These are the smallest computers designed to be used by individuals. PCs can be used for variety of tasks like documentation, calculations, illustration and entertainment. The power of network and internet has also made it more useful. Now computers are also used for communication and socialization.

Q.14: DESCRIBE THE CLASSIFICATION OF COMPUTERS ACCORDING TO PURPOSE.**CLASSIFICATION OF COMPUTERS ACCORDING TO PURPOSE:**

According to purpose, computers are either general purpose or special purpose.

(a) General Purpose Computers:

Most computers in use today are General Purpose Computers. These computers can process variety of tasks. These computers can store and execute different programs in their memory. Therefore, various tasks like word processing (typing & editing), payrolls, accounts, inventory control, manipulating facts in a database, making scientific calculations and statistical data analysis, and controlling security system of an organizations are achieved by these computers. Desktops, laptops, tablets and smart phones are examples of General Purpose Computers.

(B) SPECIAL PURPOSE COMPUTERS:

As the name states, Special Purpose Computers designed to perform specific tasks. Special Purpose Computers repeatedly perform single job more efficiently. They are also known as Dedicated Computers. These computers are useful in traffic lights control system, gravitational system aviation, weather forecasting, satellite tracking and ATMs.

Q.15: EXPLAIN THE USES OF COMPUTERS IN**ROLE AND USES OF COMPUTER:**

The role of computer in fields of lives is increasing day by day. Different aspects of the changing role of computer are as follows:

Computers in Different Fields of Life:

We are living today in the information era and the information has become one of the most valuable assets. To process this information, we make use of computers in various fields in our daily life. Computers have been integrated in our life in different shapes and sizes like desktops, laptops, mobile phones, gaming consoles and smart devices. Our lives have become so dependent of computers that we cannot work even a single day without the help of it. We have automated many of the repetitive tasks with the help of more advanced, faster and accurate computers and brought our routine tasks, office work, businesses, researches and industrial applications to a new level.

Computers are largely used in every field of life. Manufacturing, ecommerce, education, medical, banking, communication, entertainments engineering, agriculture, architecture, business, defense, and sports are highly influenced by computers.

COMPUTERS IN TODAY'S WORLD:

21st century is the era of technological revolution. In the past technology was only used to access and share information. However, with the passage of time, the rapid change in technology has made it nearly impossible to perform any human activity without its use. Computers are playing an important part in our daily life. News, weather updates, travelling information and bookings, money transfer and even taxi bookings are done with the help of mobile phone apps. Many managerial tasks are also done by computer software. School, library and hospital management are some examples. Online shopping is becoming a trend in our society too.

Computers are also providing variety of choices for entertainment like online games in which players from different parts of the world can participate simultaneously. 3D graphical software has given a new look to movies in which different comic characters can act with human actors. Photo, sound and video editor software not only help artists and singers to produce extraordinary creations but also inexperienced people can enjoy their productivity with ease of use.

Q.16: DISCUSS THE EMERGING COMPUTER TECHNOLOGY FOR TOMORROW'S WORLD.

The emerging technologies are also getting better day by day and we can imagine that how computer will be used in future. Artificial intelligence, robotics, wireless communication and virtual realities some areas which are shaping the modern world.

Artificial Intelligence:

Artificial Intelligence is based on simulating human intelligence in machines that can easily mimic and execute tasks from simple to more complex operations. The term AI may also be referred to any machine that displays qualities associated with a human brain such as learning, reasoning, and problem solving. AI is also used for machine learning. It learns from our daily routines and suggests us different options. Like Google maps suggest the best ways for our daily commute. AI is vastly used in scientific experiment, healthcare and Robots are the machines that can be controlled by a computer and are capable of carrying out a complex series of actions automatically. Robot can be controlled by an external control device or through programming. Robotics deals with designing, creating and programming of the robot and enhancing it by taking sensory feedback and processing information. Robots are commonly used in manufacturing, industry, science, medicine and education.

Wireless Communication:

Wireless Communication has become an integral part of our lives. Wireless communication technology transmits information using electromagnetic waves like IR (Infrared), RF (Radio Frequency), satellite, etc. Global Positioning System (GPS) is now used to find real time location of people and objects. Smart phones that include 3G, 4G and 5G networks have transformed the conventional use of telephones. These phones not only provide better coverage but also fast mobile internet experience. Wireless Communication is also giving new meaning to the socialization and human interactivity.

Virtual Reality:

Virtual Reality is an artificial environment that is created with software and presented to the user in a way that the user feels it as a real environment. Virtual reality can be used for the simulation of a real environment for training and education. It is also used for development of an imagined environment for a game or interactive story.

Q.17: DISCUSS THE SCOPE OF THE DIFFERENT CAREERS IN IT**CAREERS IN IT FIELD:**

Today Computer Science plays a key role in all fields of life. Computer Science (CS) jobs are ranked as having some of the highest salaries in the world. It's no surprise that computer science and information technology (IT) are changing tomorrow's job market. Let's look at different careers that are present in the IT sector.

(I) SOFTWARE ENGINEER:

A Software Engineer is a person who uses different programming languages to develop software products like games, learning management system (LMS), business applications, educational and entertainment software.

(II) WORK ADMINISTRATOR:

Network Administrator is an IT expert who manages an organization's network. He or She is responsible for installing, maintaining and upgrading any software or hardware required to efficiently run a computer network.

(III) DATABASE ADMINISTRATOR:

A Database Administrator (also known as DBA) is a skilled professional who maintains a secure database environment in an office, business or organization.

(IV) WEB DESIGNER:

Web is short for World Wide Web or www. This is another name for internet. Web designers are people who build websites. They prepare a site's content with eye catching designs, attractive images and strong text. They rely on many web designing tools to ensure an interactive website. How the site works and how it looks is the responsibility of web designers. They are also responsible for maintaining and updating an existing site.

(V) GRAPHIC DESIGNER:

Graphic is another word for image. Graphics Designers develop overall layouts of brochures, magazines and other types of published advertisements and documents. They do this by using various computer graphics software. This task is usually accomplished by combining art and technology, conveying ideas through images, layout of websites and printed matter.

(VI) INFORMATION SECURITY ANALYST:

These people protect networks... They plan carry out security measures to ensure that no loss of data (or information) occurs. We can call them the police force of the internet. Computer technology has expanded into online banking and businesses. Computer hackers are always on the lookout for customer account details that they can use to I money. Information Security Analysts ensure that no such theft occurs.

(VII) COMPUTER SCIENCE OR IT TEACHER:

Another very rewarding career for Computer Science graduates and post-graduates is teaching in schools, colleges and universities. These instructors instill in their students the importance of computers in today's world, and the impact these will have in their later lives.

Q.18: DESCRIBE COMPUTER HARDWARE AND THEIR FUNCTIONS.**COMPUTER HARDWARE:**

Computers consist of software and hardware. Software is the programs and applications that run on computer. Hardware is the physical parts of the computer that run programs and applications. Hardware can be seen and touched while software

cannot. Computer hardware is a collection of all the physical parts or components of a computer. It includes the computer casing, the CPU or Central Processing Unit, computer memory, VGA graphics card, sound card, and the motherboard.

System Unit:

A System Unit is the part of a computer that contains the primary devices. The system unit performs operations and produces results for complex calculations. It includes the motherboard, CPU, RAM and other components, as well as the case in which these devices are enclosed.

(I) Motherboard:

The motherboard is the main board which connects different parts of computer. It includes the following general components: Microprocessor (CPU), Slots, Ports, Buses, RAM, ROM and other electronic components for example resistors, capacitors, diodes, transistors, jumpers etc. Just like nervous system it allows communication between all parts of the computer. We can find CPU, memory slots, expansion slots and a number of chipsets on motherboard. Motherboards are made of layered fiberglass. These layers contain copper lines that form the circuitry by which electrical signals are provided to all parts of computer. Various components of computer may be attached, detached and upgraded on motherboard according to need of the user. Motherboards also have connectors called ports. These ports are used to connect input, output and other peripheral devices.

(II) MICROPROCESSOR (CPU):

CPU or microprocessor is the brain of computer. The microprocessor is a chip containing millions of tiny transistors. These transistors manipulate data. Microprocessor performs all the calculations necessary to make the computer work. These calculations are performed at very high speed and accuracy. Microprocessor is made up of silicon. The microprocessor fetches, decodes, executes and stores all the instructions given by the user or any other device. The speed of computer mainly depends upon the speed of microprocessor. A microprocessor that has faster clock speed (measured in GHz) and more cores and cache works faster.

Q.19: LIST ANY FIVE COMPONENTS PRESENT ON MOTHERBOARD.**LIST OF COMPONENTS PRESENT ON MOTHERBOARD:**

Microprocessor (CPU), Slots, Ports, Buses, RAM, ROM and other electronic components for example resistors, capacitors, diodes, transistors, jumpers etc.

Q.20: WRITE A NOTE ON INPUT DEVICES.**INPUT DEVICES:**

Input devices are used to enter data into computers. These devices can be Text Entering, Pointing and Image Scanning Devices.

(I) TEXT ENTERING DEVICES:**KEYBOARD:**

It is the most common text entering device and used to enter data usually in text format or to perform other controlling functions. When a key is pressed, keyboard controller chip sends its corresponding code in keyboard buffer called scan code, which is processed by CPU.

(II) POINTING DEVICES (MOUSE AND ITS VARIANTS):

They are used for the quick movement of cursor on screen needed usually in graphic mode. It includes mouse, joystick, track ball and track pad.

(III) IMAGE SCANNERS:

Scanners convert image into electronic format understandable by computers through light sensing. They also work on Optical Recognition which occurs when a device scans a clear printed surface and translates the image into machine readable formats that a computer understands. Image scanners include Optical Mark Recognition (OMR), Optical Barcode Reader (OBR) and Optical Character Recognition (OCR).

(IV) OTHERS TYPES OF INPUT DEVICES:

There are also other input devices. For example, microphones are used to accept sound input and convert it in digital audio format. Microphones are also used for voice recognition which can convert voice input into text files. Touch Screen is also used for input. It accepts input directly on monitor by touching finger or any object on the screen. Magnetic Ink Character Recognition (MICR) and Magnetic Strip Reader are also used for input.

Q.20: WHAT DO YOU KNOW ABOUT OUTPUT DEVICES?**OUTPUT DEVICES:**

A hardware device that sends data from a computer (CPU) to another device or user is called an output device. The most common output devices of a computer are monitor and printer. There are two types of output devices.

(I) SOFT COPY OUTPUT DEVICES:

It is screen display or voice output. It is volatile output and lost when other output is shown or computer is turned off. Following are some of the devices used to give output in soft form.

(A) MONITORS:

It is TV like device that displays data by small Monitors are of two types.

- Cathode Ray Tube (CRT)
- Flat Panel Display (FDP)

(B) DATA PROJECTORS:

Data projectors are used to show colorful slides and images directly from computer disk on a wall or large screen through an optical lens. They are also called digital light projectors and video projectors.

© SPEAKERS:

Speakers give output disabilities form of sound. They are good for people with visual disabilities and where display is not easy.

(II) HARD COPY DEVICES:

It gives output on paper. It is nonvolatile output that is relatively stable and permanent form. Hard Copy Output Devices are:

(A) PRINTERS:

They usually give output on paper and can print both text and graphics. There are two types of printers:

Impact, Non-impact

(B) PLOTTERS:

Like printer it gives images on paper but typically used to print large format images such as maps, construction drawing, advertising hoardings etc.

Q.21: WHAT ARE BUSES IN COMPUTER? DEFINE THE TYPES OF BUSES.**BUSES:**

In computer, Buses are the electric paths on which data is sent and received by different components. They are just like roads. As roads connect different places, buses connect all the parts of the computer to each other. They also connect all internal components on the motherboard. There are three types of buses:

(i) Control bus

(ii) Data bus

(iii) Address bus

(i) CONTROL BUS:

control Bus carries command between different components to control computer.

(ii) DATA BUS:

It carries data between the processor, memory unit and other components.

(III) ADDRESS BUS:

It carries the address of the data (but not the data).

address bus is used to specify memory location to be used by micro-process for specific operation.

Q.22: WHICH MONITOR WILL YOU PREFER IN YOUR SCHOOL: CRT OR FPD? WHY?

I prefer FPD (Flat Panel Display) because

- (i) they take up less desk space and their prices continue to drop.
- (ii) FPDs are more susceptible to accidental or deliberate damage
- (iii) FPDs have more lifetime than CRT.
- (iv) FPDs are, on average, brighter than CRTs.
- (v) An inexpensive FPD may have a better contrast ratio than an excellent CRT monitor.
- (vi) Those big CRT monitors also produce a lot of heat, which wastes energy.
- (vii) FPDs consume less power and this can lead to a considerable reduction in the bill.

Q.23: DEFINE COMPUTER SOFTWARE. WRITE THE NAMES OF ITS TYPES.**COMPUTER SOFTWARE:**

Software is a set of instructions that a computer uses to perform a task. It is a general term for a computer program. There are two main types of software:

(I) SYSTEM SOFTWARE**(II) APPLICATION SOFTWARE****Q.24: DISCUSS IMPACT AND NON-IMPACT PRINTERS WITH EXAMPLES.**

Impact and Non-Impact Printers are two categories of the printer. Impact printers involve mechanical components for conducting printing. While in Non-Impact printers, no mechanical moving component is used.

Impact Printers:

It is a type of printer that works by direct contact of an ink ribbon with paper. These printers are typically loud but remain in use today because of their unique ability to function with multi part forms. An impact printer has mechanisms resembling those of a typewriter. Examples of Impact Printers: Dot-matrix printers, Daisy-wheel printers, and line printers.

Non-Impact Printers:

It is a type of printer that does not hit or impact a ribbon to print. They used laser, xerographic, electrostatic, chemical and inkjet technologies. Non-impact printers are generally much quieter. They are less likely to need maintenance or repairs than earlier impact printers.

Example of Non-Impact Printers: Inkjet printers and Laser printers.

Q.25: DESCRIBE STORAGE DEVICES AND ITS TYPES.**STORAGE DEVICES:**

A storage device refers to a hardware used to store information. There are two types of storage devices: Primary and Secondary.

(i) Primary Storage Devices:

Primary storage devices are used by computer during processing. They are quite smaller in storage capacity. Most primary storage devices are found inside the computer, and they have the fastest access to data. Primary data include RAM and ROM.

ROM: ROM is the Read Only Memory. It is permanent memory. ROM is quite small in capacity.

It stores the major setting of computer permanently.

RAM: RAM is Random Access Memory. RAM is volatile means it loses its content as the power supply is disconnected. This is used to store data and instructions temporarily.

(ii) Secondary Storage Devices:

Secondary storage devices have a larger storage capacity and can store data permanently. Users

save their data on secondary storage devices. Hard Disk, CD and DVD, SD Card and USB flash

disk are examples of secondary storage devices. Example: Hard Disc, Secure Digital Card, Pen

Drive, DVD, CD,

Q.26: DESCRIBE THE BASIC OPERATIONS OF A COMPUTER.**BASIC OPERATIONS OF A COMPUTER:**

A computer is a machine that acts according to the instructions given by the user. A computer

performs four basic operations: input, processing, output and storage.

(i) Input:

Computer input has many forms. It can be from a command entered using a keyboard or a mouse. It can be data sent from another computer on a net-work.

(ii) Processing:

Processing is done inside the computer by CPU. Processing is the conversion of input into output. After processing, data is turned into meaningful information. This is carried out with the help of arithmetic and logical operation

(iii) Storage:

Storage refers to the holding or saving data. RAM, ROM and Hard Disk are the devices that are used for storage.

(B) COMPILER:

It translates the entire high-level language program at once into machine language before it is executed.

c) Interpreter:

It translates the high-level language program line by line into machine language.

Q.27: DESCRIBE APPLICATION SOFTWARE.**APPLICATION SOFTWARE:**

Application Software is used to complete specific tasks, such as creating documents, databases, spreadsheets and presentations. Computer games, media players and web browsers are also the examples of application software. User needs to install application software to accomplish specific tasks. Application software are not pre-installed on operating system. They are installed separately. For example, making spreadsheet is not possible with Windows. For that application software such as MS Excel is used. There are different types of application software.

(i) Productivity Software:

This kind of application software is used to produce things such as documents, spreadsheets, databases and presentations. Many of the productivity software are intended to be used in businesses and offices. MS Office for windows is an example of productivity software.

(ii) Business Software:

Business Software are used to manage business activities efficiently. It is used to provide many business functions such as billing as billing database management and inventory management.

(iii) Entertainment Software:

This type of software provides amusement and fulfills a user's hobby. The most common entertainment software are video games.

(iv) Educational Software:

Educational software is used for teaching and learning. This kind of software is used in many schools to enhance knowledge of students in learning different subjects.

Q.25: DESCRIBE STORAGE DEVICES AND ITS TYPES.**STORAGE DEVICES:**

A storage device refers to a hardware used to store information. There are two types of storage devices: Primary and Secondary.

(i) Primary Storage Devices:

Primary storage devices are used by computer during processing. They are quite smaller in storage capacity. Most primary storage devices are found inside the computer, and they have the fastest access to data. Primary data include RAM and ROM.

ROM: ROM is the Read Only Memory. It is permanent memory. ROM is quite small in capacity.

It stores the major setting of computer permanently.

RAM: RAM is Random Access Memory. RAM is volatile means it loses its content as the power supply is disconnected. This is used to store data and instructions temporarily.

(ii) Secondary Storage Devices:

Secondary storage devices have a larger storage capacity and can store data permanently. Users

save their data on secondary storage devices. Hard Disk, CD and DVD, SD Card and USB flash

disk are examples of secondary storage devices. Example: Hard Disc, Secure Digital Card, Pen

Drive, DVD, CD,

Q.26: DESCRIBE THE BASIC OPERATIONS OF A COMPUTER.**BASIC OPERATIONS OF A COMPUTER:**

A computer is a machine that acts according to the instructions given by the user. A computer

performs four basic operations: input, processing, output and storage.

(i) Input:

Computer input has many forms. It can be from a command entered using a keyboard or a mouse. It can be data sent from another computer on a net-work.

(ii) Processing:

Processing is done inside the computer by CPU. Processing is the conversion of input into output. After processing, data is turned into meaningful information. This is carried out with the help of arithmetic and logical operation

(iii) Storage:

Storage refers to the holding or saving data. RAM, ROM and Hard Disk are the devices that are used for storage.